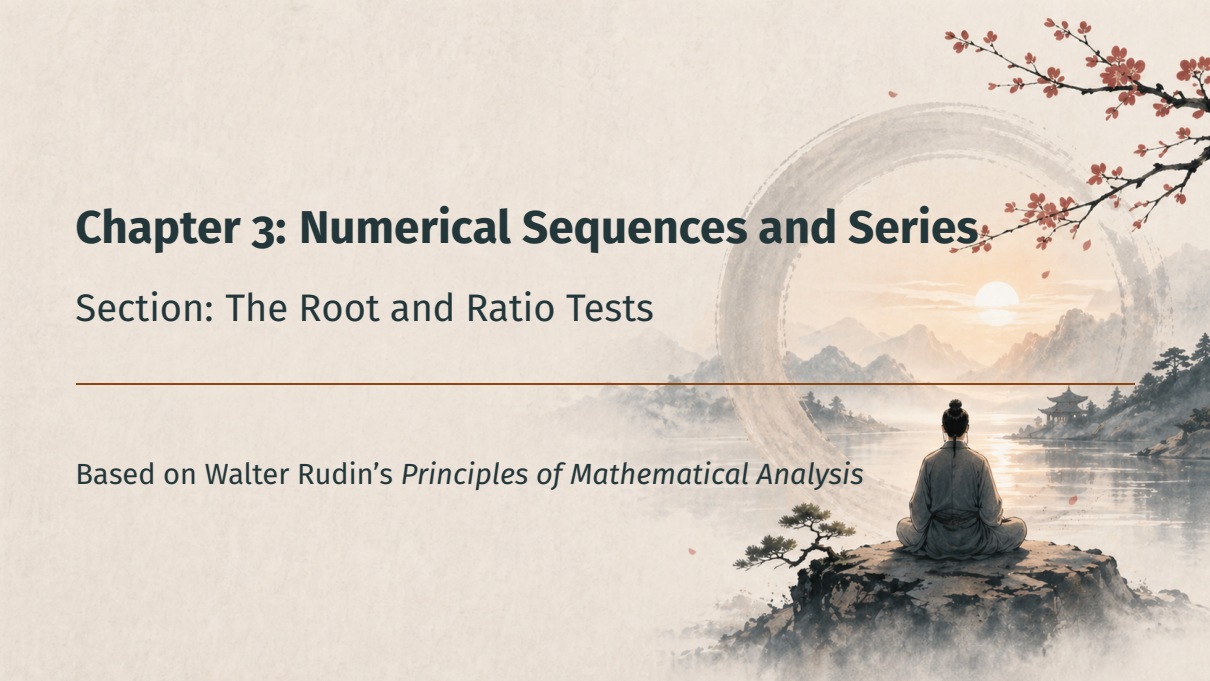


Chapter 3: Numerical Sequences and Series

The background of the slide is a traditional Chinese ink wash painting. It depicts a person with their hair in a bun, wearing a grey robe, sitting in a meditative lotus position on a dark, craggy rock. The person is facing away from the viewer, looking out over a calm body of water. In the distance, there are misty mountains and a small pavilion with a traditional Chinese roof on the right bank. A large, faint, circular frame, resembling a ring or a portal, is centered in the background, framing the sunset. The sun is a bright, glowing orb on the horizon. In the upper right corner, there are branches of cherry blossoms with small red flowers. The overall atmosphere is peaceful and contemplative.

Section: The Root and Ratio Tests

Based on Walter Rudin's *Principles of Mathematical Analysis*

Outline

Motivation

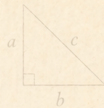
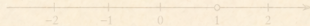
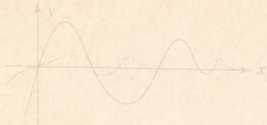
The Root Test

The Ratio Test

Examples and Remarks

Root vs. Ratio

Summary



$$a^2 + b^2 = c^2$$

Motivation



$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

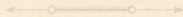


Why We Want New Tests

- The comparison test needs a **known series** to compare against



$$f(x) = \sin x$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Why We Want New Tests

- The comparison test needs a **known series** to compare against
- We want **intrinsic** tests: use only the terms a_n themselves



$$f(x) = \sin x$$



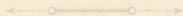
$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Why We Want New Tests

- The comparison test needs a **known series** to compare against
- We want **intrinsic** tests: use only the terms a_n themselves
- Both new tests secretly compare $\sum a_n$ to a **geometric series**



$$f(x) = \sin x$$



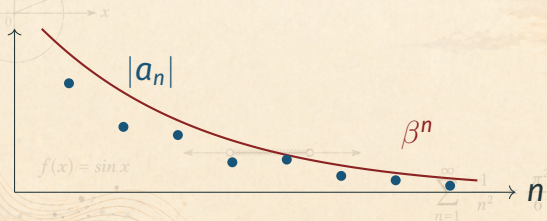
$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

The Benchmark: The Geometric Series

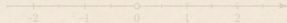
Recall

The geometric series $\sum_{n=0}^{\infty} x^n$ converges $\iff |x| < 1$.

Key idea: if terms are eventually dominated by β^n with $\beta < 1$, convergence follows by **comparison**.



The Root Test



$$x^2 + y^2 = r^2$$



$$a^2 + b^2 = c^2$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$



Theorem 3.33: The Root Test

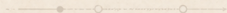
Theorem 3.33 (Root Test)

Given $\sum a_n$, put

$$\alpha = \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}.$$



$$a^2 + b^2 = c^2$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

Theorem 3.33: The Root Test

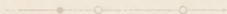
Theorem 3.33 (Root Test)

Given $\sum a_n$, put $\alpha = \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}$. Then

- (a) if $\alpha < 1$, $\sum a_n$ **converges**;
- (b) if $\alpha > 1$, $\sum a_n$ **diverges**;
- (c) if $\alpha = 1$, the test gives **no information**.



$$a^2 + b^2 = c^2$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

Theorem 3.33: Three Regions for α



Proof of Theorem 3.33 — Strategy

Goal: Decide convergence of $\sum a_n$ from α alone.

Approach — three cases:

1. $\alpha < 1$: trap each term below β^n for a fixed $\beta < 1$, then compare.
2. $\alpha > 1$: find infinitely many terms with $|a_n| > 1$, so $a_n \not\rightarrow 0$.
3. $\alpha = 1$: exhibit one convergent and one divergent example.

Proof of Theorem 3.33 — Case (a): $\alpha < 1$

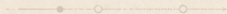
Case (a): $\alpha < 1$

Choose β with $\alpha < \beta < 1$. By Theorem 3.17(b), there is an integer N such that

$$\sqrt[n]{|a_n|} < \beta \quad (n \geq N).$$



$$a^2 + b^2 = c^2$$



$$\sum_{n=1}^{\infty} \frac{1}{n^2}$$

Proof of Theorem 3.33 — Case (a): $\alpha < 1$

Case (a): $\alpha < 1$

Choose β with $\alpha < \beta < 1$, and N with $\sqrt[n]{|a_n|} < \beta$ for $n \geq N$. Raise to the n th power:

$$|a_n| < \beta^n \quad (n \geq N).$$

Since $0 < \beta < 1$, $\sum \beta^n$ converges. By the comparison test, $\sum a_n$ converges.

Proof of Theorem 3.33 — Case (b): $\alpha > 1$

Case (b): $\alpha > 1$

By Theorem 3.17, there is a subsequence with

$$\sqrt[n_k]{|a_{n_k}|} \rightarrow \alpha > 1.$$

Hence $|a_n| > 1$ for **infinitely many** n .

So $a_n \not\rightarrow 0$. By Theorem 3.23, $\sum a_n$ diverges.

Proof of Theorem 3.33 — Case (c): $\alpha = 1$

Case (c): $\alpha = 1$: no information

Consider

$$\sum \frac{1}{n} \quad (\text{diverges}), \quad \sum \frac{1}{n^2} \quad (\text{converges}).$$

For *both*, $\alpha = \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|} = 1$ (since $\sqrt[n]{n} \rightarrow 1$).

Same α , opposite behavior $\Rightarrow$ the test cannot decide.

The Ratio Test

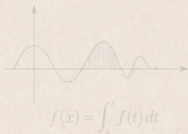


Theorem 3.34: The Ratio Test

Theorem 3.34 (Ratio Test)

The series $\sum a_n$

(a) **converges** if $\limsup_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$;



Theorem 3.34: The Ratio Test

Theorem 3.34 (Ratio Test)

The series $\sum a_n$

(a) **converges** if $\limsup_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| < 1$;

(b) **diverges** if $\left| \frac{a_{n+1}}{a_n} \right| \geq 1$ for all $n \geq n_0$, some fixed integer n_0 .

Theorem 3.34: A Warning

Note

Knowing $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1$ implies **nothing** about convergence.

Once again $\sum \frac{1}{n}$ and $\sum \frac{1}{n^2}$ both have ratio limit 1, yet one diverges and the other converges.

Proof of Theorem 3.34 — Case (a): convergence

Case (a): $\limsup |a_{n+1}/a_n| < 1$

Pick $\beta < 1$ and N with $\left| \frac{a_{n+1}}{a_n} \right| < \beta$ for $n \geq N$. Cascading:

$$|a_{N+1}| < \beta |a_N|, \quad |a_{N+2}| < \beta^2 |a_N|, \quad \dots, \quad |a_{N+p}| < \beta^p |a_N|.$$

Proof of Theorem 3.34 — Case (a): convergence

Case (a): $\limsup |a_{n+1}/a_n| < 1$

With $n > N$, the cascade becomes

$$|a_n| < |a_N| \beta^{-N} \cdot \beta^n \quad (n > N).$$

The factor $|a_N| \beta^{-N}$ is a **constant**. Since $\sum \beta^n$ converges, $\boxed{\sum a_n \text{ converges}}$ by comparison.

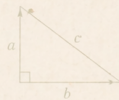
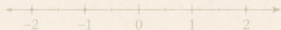
Proof of Theorem 3.34 — Case (b): divergence

Case (b): $|a_{n+1}| \geq |a_n|$ for $n \geq n_0$

The sizes of the terms never decrease past n_0 :

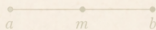
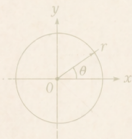
$$|a_{n_0}| \leq |a_{n_0+1}| \leq |a_{n_0+2}| \leq \cdots$$

Hence $a_n \not\rightarrow 0$, so $\sum a_n$ diverges.



$$a^2 + b^2 = c^2$$

Examples and Remarks



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

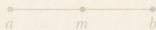
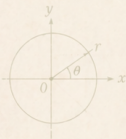


Example 3.35(a): Root Wins, Ratio Fails

Example 3.35(a)

$$\frac{1}{2} + \frac{1}{3} + \frac{1}{2^2} + \frac{1}{3^2} + \frac{1}{2^3} + \frac{1}{3^3} + \frac{1}{2^4} + \frac{1}{3^4} + \cdots$$

Two geometric series — powers of $\frac{1}{2}$ and powers of $\frac{1}{3}$ — **interleaved**.



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$a^2 + b^2 = c^2$$

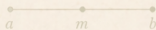
Example 3.35(a): Root Wins, Ratio Fails

Example 3.35(a)

$$\begin{aligned}\liminf \frac{a_{n+1}}{a_n} &= 0, & \limsup \frac{a_{n+1}}{a_n} &= +\infty, \\ \liminf \sqrt[n]{a_n} &= \frac{1}{\sqrt{3}}, & \limsup \sqrt[n]{a_n} &= \frac{1}{\sqrt{2}}.\end{aligned}$$

Root test: $\alpha = 1/\sqrt{2} < 1 \Rightarrow$ **converges.**

Ratio test: $\limsup = +\infty \Rightarrow$ **no conclusion.**


$$a \quad m \quad b$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Example 3.35(b): Same Lesson

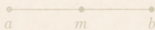
Example 3.35(b)

$$\frac{1}{2} + 1 + \frac{1}{8} + \frac{1}{4} + \frac{1}{32} + \frac{1}{16} + \frac{1}{128} + \frac{1}{64} + \dots$$

$$\liminf \frac{a_{n+1}}{a_n} = \frac{1}{8}, \quad \limsup \frac{a_{n+1}}{a_n} = 2, \quad \lim \sqrt[n]{a_n} = \frac{1}{2}.$$

Root test: $\frac{1}{2} < 1 \Rightarrow$ converges.

Ratio test: $\limsup = 2 > 1 \Rightarrow$ no conclusion.


$$a \quad m \quad b$$

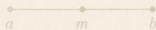
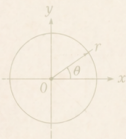
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

Remark 3.36: Comparing the Two Tests

Remark 3.36

The **ratio test** is frequently *easier to apply* — computing ratios is usually simpler than computing n th roots.



$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$a^2 + b^2 = c^2$$

Remark 3.36: Comparing the Two Tests

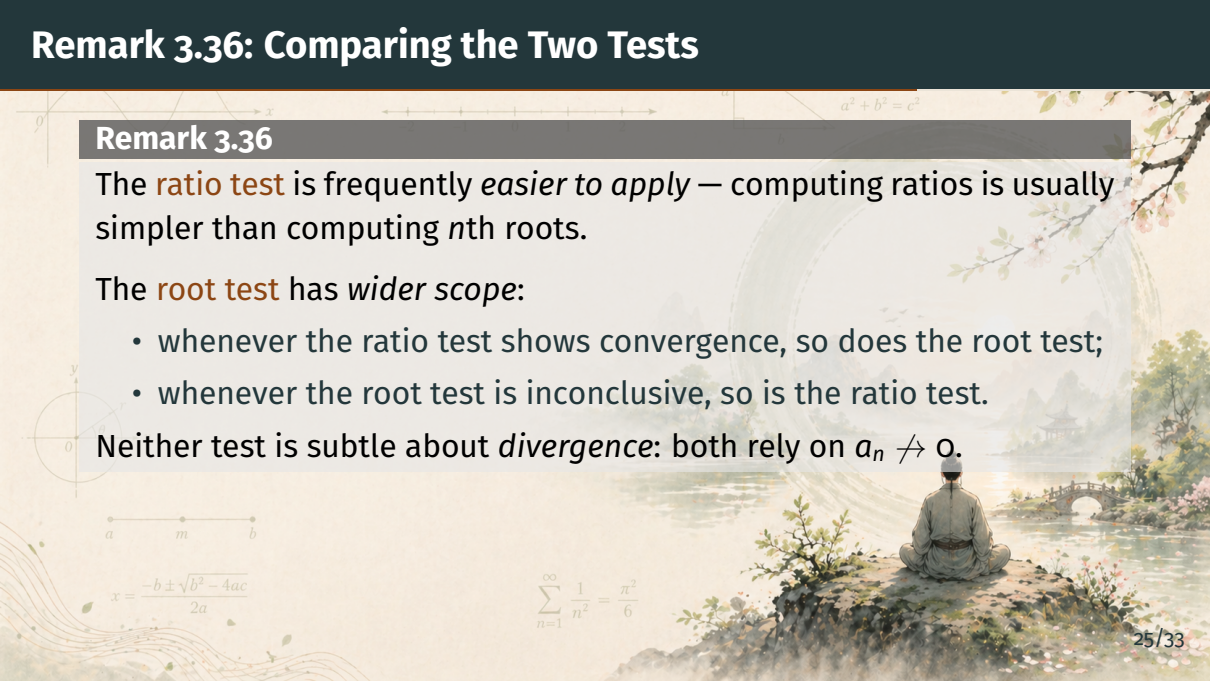
Remark 3.36

The **ratio test** is frequently *easier to apply* — computing ratios is usually simpler than computing n th roots.

The **root test** has *wider scope*:

- whenever the ratio test shows convergence, so does the root test;
- whenever the root test is inconclusive, so is the ratio test.

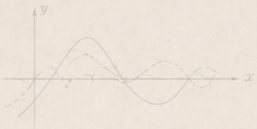
Neither test is subtle about *divergence*: both rely on $a_n \not\rightarrow 0$.


$$a \quad m \quad b$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$a^2 + b^2 = c^2$$

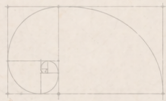


$$e^{i\theta} = \cos \theta + i \sin \theta$$



$$\nabla^2 \phi = 0$$

Root vs. Ratio



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$



Theorem 3.37: Roots Squeezed Between Ratios

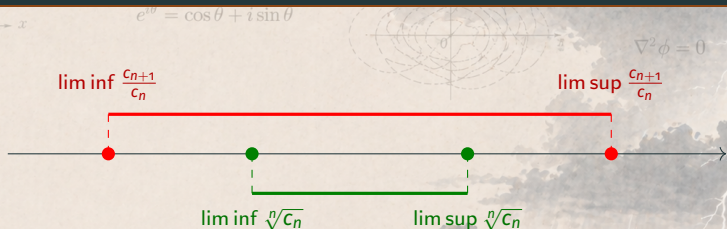
Theorem 3.37

For any sequence $\{c_n\}$ of *positive* numbers,

$$\liminf_{n \rightarrow \infty} \frac{c_{n+1}}{c_n} \leq \liminf_{n \rightarrow \infty} \sqrt[n]{c_n},$$

$$\limsup_{n \rightarrow \infty} \sqrt[n]{c_n} \leq \limsup_{n \rightarrow \infty} \frac{c_{n+1}}{c_n}.$$

Theorem 3.37: The Squeeze, Pictured



The **root** interval (green) sits *inside* the **ratio** interval (red).

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

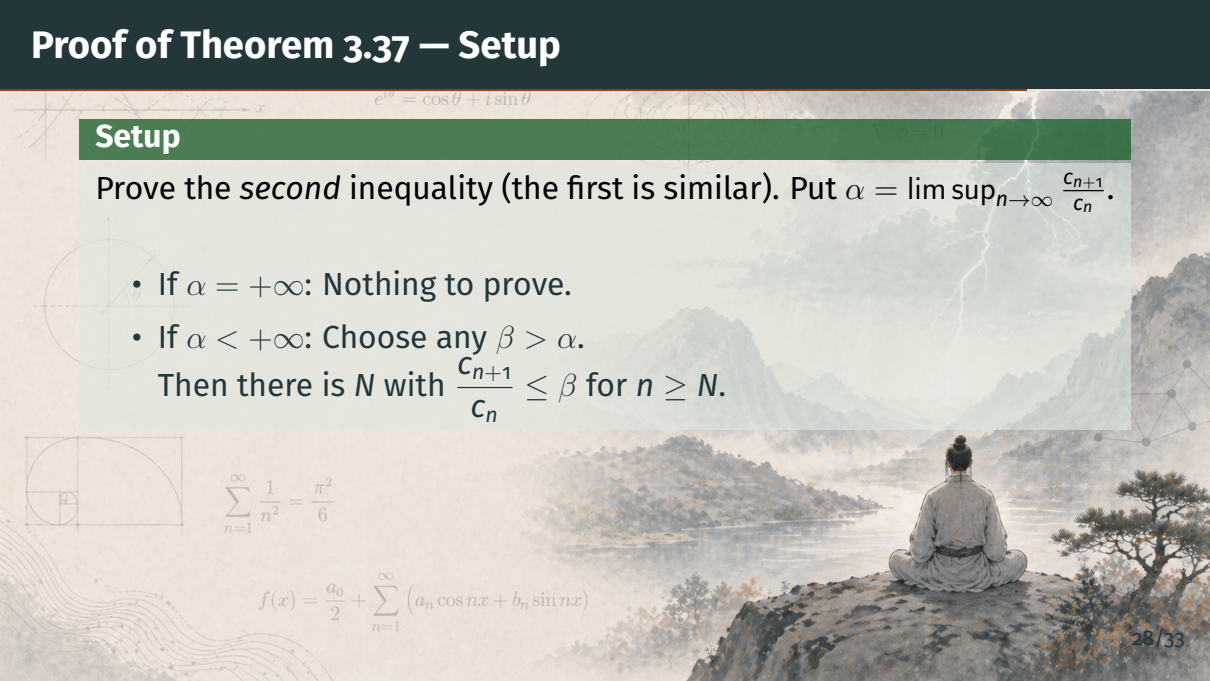
$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Proof of Theorem 3.37 — Setup

Setup

Prove the *second* inequality (the first is similar). Put $\alpha = \limsup_{n \rightarrow \infty} \frac{c_{n+1}}{c_n}$.

- If $\alpha = +\infty$: Nothing to prove.
- If $\alpha < +\infty$: Choose any $\beta > \alpha$.
Then there is N with $\frac{c_{n+1}}{c_n} \leq \beta$ for $n \geq N$.


$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$

Proof of Theorem 3.37 — The Cascade

Cascade

For $k = 0, 1, \dots, p - 1$: $c_{N+k+1} \leq \beta c_{N+k}$. Multiplying all of these:

$$c_{N+p} \leq \beta^p c_N, \quad \text{i.e.} \quad c_n \leq c_N \beta^{-N} \cdot \beta^n \quad (n \geq N).$$

Proof of Theorem 3.37 — Conclusion

Conclusion

Take n th roots: $\sqrt[n]{c_n} \leq \sqrt[n]{c_N} \beta^{-N} \cdot \beta.$

Since $\sqrt[n]{c_N} \beta^{-N} \rightarrow 1$, by Theorem 3.20(b)

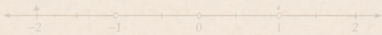
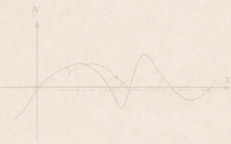
$$\limsup_{n \rightarrow \infty} \sqrt[n]{c_n} \leq \beta. \quad (18)$$

True for every $\beta > \alpha$, hence

$$\boxed{\limsup_{n \rightarrow \infty} \sqrt[n]{c_n} \leq \alpha.}$$

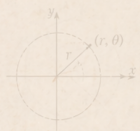
$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} (a_n \cos nx + b_n \sin nx)$$



$$a^2 + b^2 = c^2$$

Summary



$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Summary

Key takeaways:

- **Root test (3.33):** with $\alpha = \limsup \sqrt[n]{|a_n|}$: converges if $\alpha < 1$, diverges if $\alpha > 1$, inconclusive if $\alpha = 1$.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Summary

Key takeaways:

- **Root test (3.33):** with $\alpha = \limsup \sqrt[n]{|a_n|}$: converges if $\alpha < 1$, diverges if $\alpha > 1$, inconclusive if $\alpha = 1$.
- **Ratio test (3.34):** converges if $\limsup |a_{n+1}/a_n| < 1$, diverges if $|a_{n+1}/a_n| \geq 1$ eventually.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$



Summary

Key takeaways:

- **Root test (3.33):** with $\alpha = \limsup \sqrt[n]{|a_n|}$: converges if $\alpha < 1$, diverges if $\alpha > 1$, inconclusive if $\alpha = 1$.
- **Ratio test (3.34):** converges if $\limsup |a_{n+1}/a_n| < 1$, diverges if $|a_{n+1}/a_n| \geq 1$ eventually.
- **Theorem 3.37:** roots are squeezed between ratios — the root test has *wider scope*.

Coming up next: Power Series.

$$\sum_{n=1}^{\infty} \frac{1}{n^2} = \frac{\pi^2}{6}$$

